

Differentiable functions – 3

Convexity and differential relations

1 сентября 2026 года

Theoretical minimum.

Def. f is convex on an interval if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$; for a differentiable f this is equivalent to the monotonicity of f' , and for a twice differentiable one to $f'' \geq 0$.

Theorem 1 (regularity of convex functions). A convex function on an open interval is continuous, has one-sided derivatives at every point, and is differentiable outside an at most countable set; its graph lies above every tangent line and below every chord.

Theorem 2 (logarithmic convexity). $f > 0$ is logarithmically convex if $\ln f$ is convex, which for a twice differentiable f means $f''f \geq (f')^2$. Logarithmically convex functions are convex, and the class is closed under sums and products.

Fact 1. Every C^2 function on a segment is a difference of two convex functions: write $f = (f + Cx^2) - Cx^2$ with $2C \geq \max |f''|$. The same is true for $\sin x$ on the whole line, which is Problem 3.

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1. (IMC 2020) Find all twice continuously differentiable $f: \mathbb{R} \rightarrow (0, +\infty)$ satisfying $f''(x)f(x) \geq 2(f'(x))^2$ for all $x \in \mathbb{R}$.
 2. Show that if A, B, C are the angles of a triangle, then $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.
 3. (VJIMC 1992) Prove that there exist two real convex functions f, g with $f(x) - g(x) = \sin x$ for all $x \in \mathbb{R}$.
 4. (RMO 2005) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be convex.

- a) Prove that f is continuous.
- b) Prove that there is a unique $g: [0, \infty) \rightarrow \mathbb{R}$ with $f(x + g(x)) = f(g(x)) - g(x)$ for all $x \geq 0$.

5. (RMO 2013)

- a) Let $f: [0, \infty) \rightarrow [0, \infty)$ be differentiable and convex with $f(x) \leq x$ for every $x \geq 0$. Show that $f'(x) \leq 1$ for every $x \geq 0$.
- b) Determine all differentiable convex $f: [0, \infty) \rightarrow [0, \infty)$ with $f(0) = 0$ and $f'(x)f(f(x)) = x$ for every $x \geq 0$.